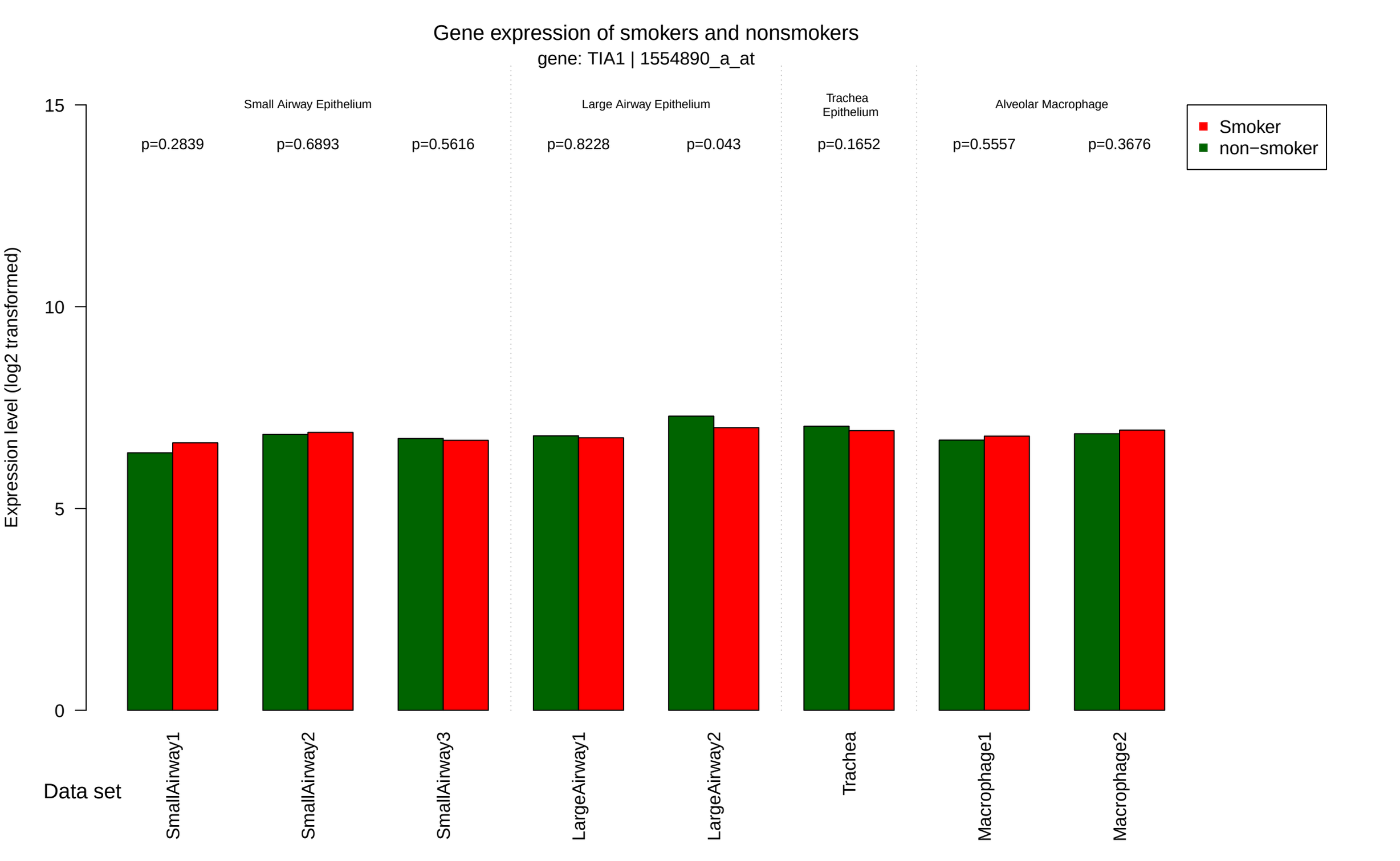
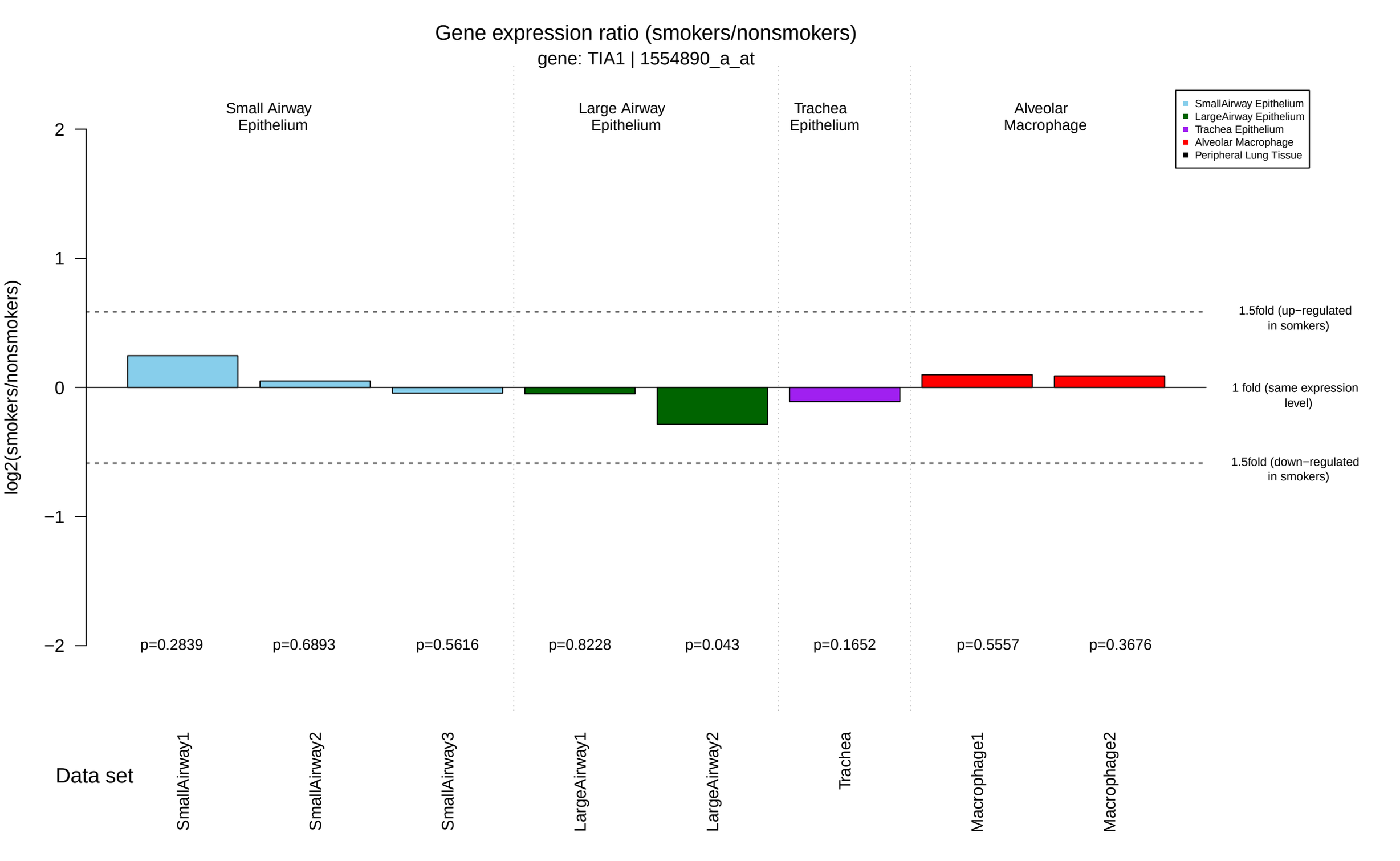
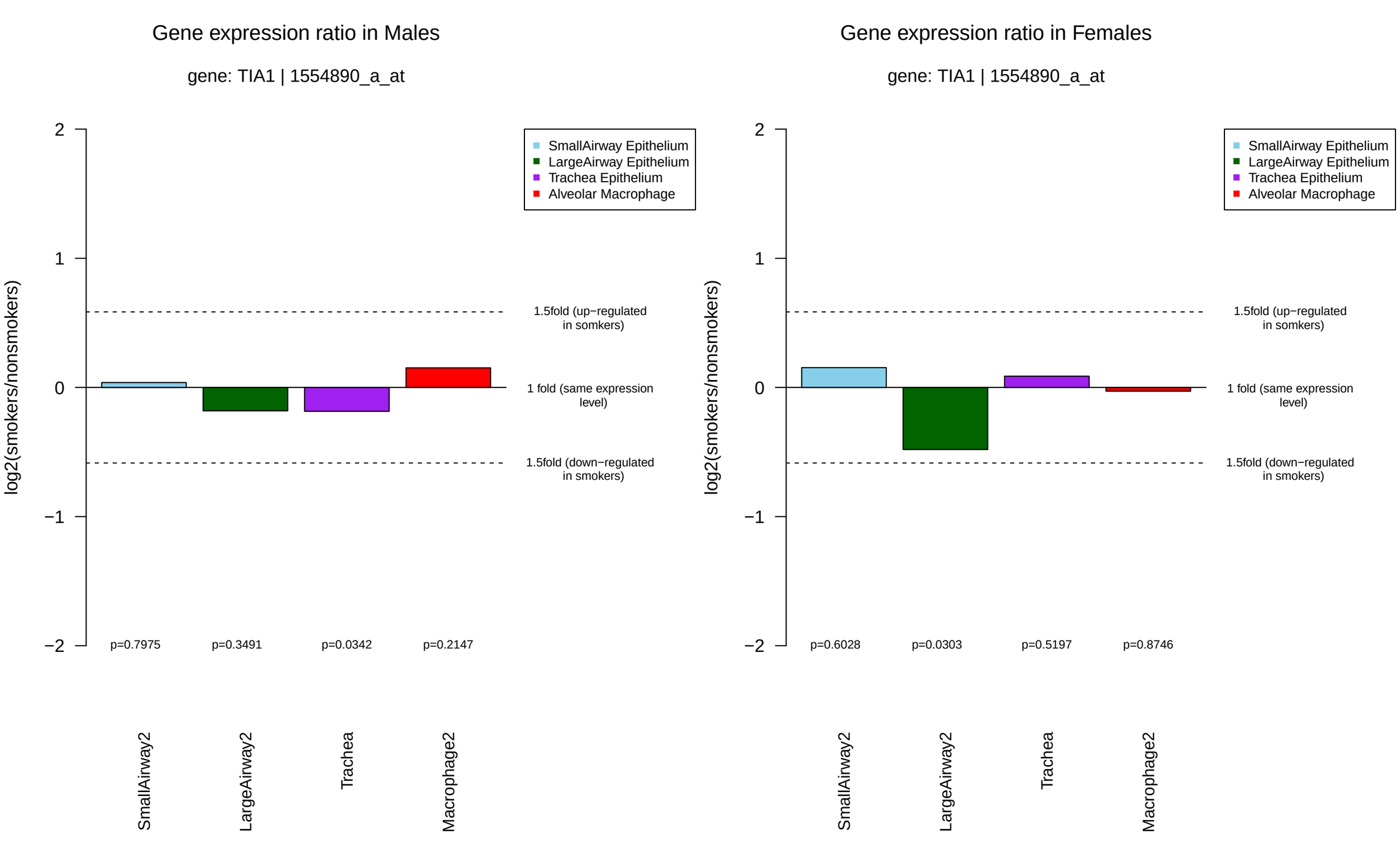
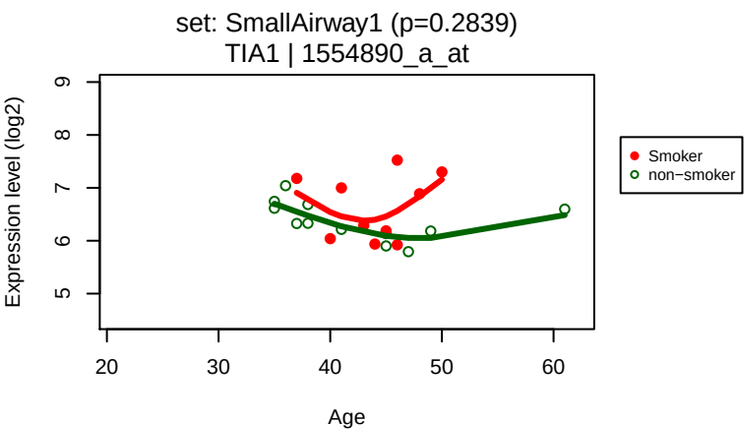
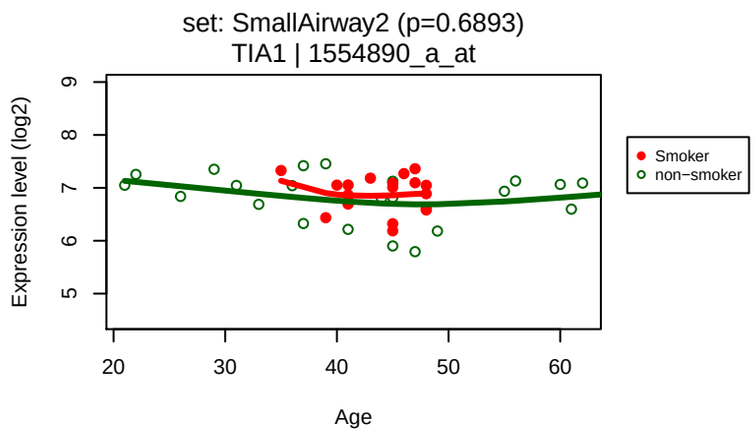
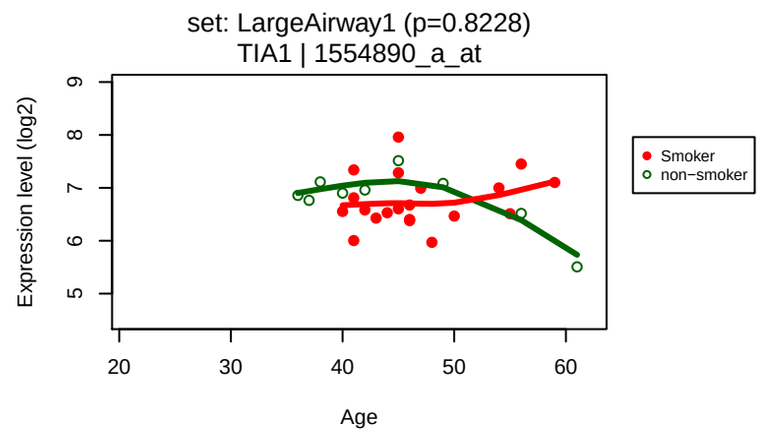

Gene expression of smokers and nonsmokers
gene: TIA1 | 1554890_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2839
p=0.6893
p=0.5616
p=0.8228
p=0.043
p=0.1652
p=0.5557
p=0.3676
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

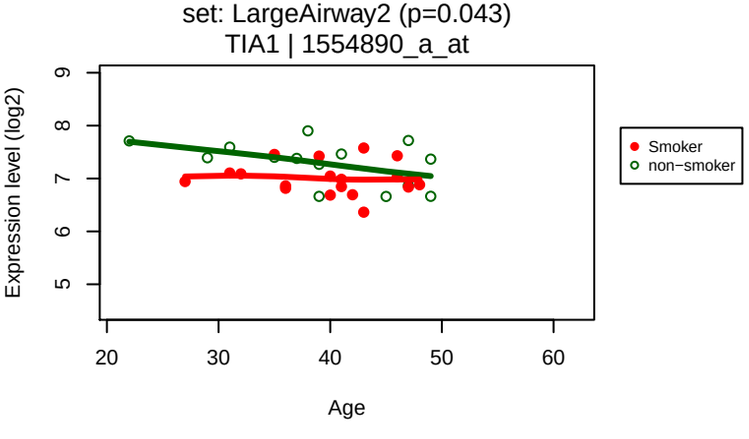

Gene expression ratio (smokers/nonsmokers)
gene: TIA1 | 1554890_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2839
p=0.6893
p=0.5616
p=0.8228
p=0.043
p=0.1652
p=0.5557
p=0.3676
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

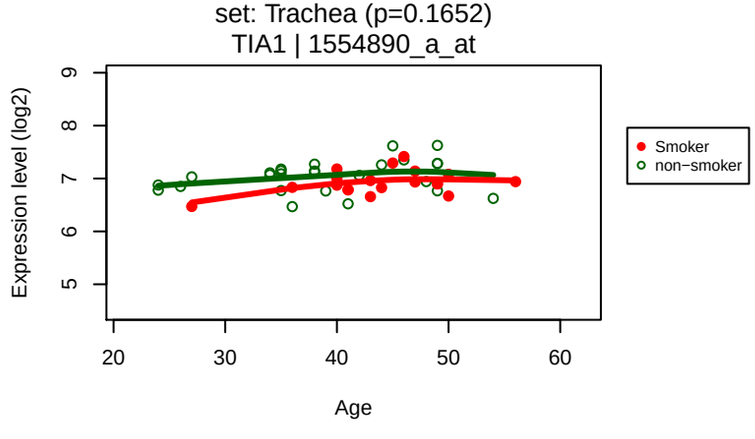

Gene expression ratio in Males
gene: TIA1 | 1554890_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7975
p=0.3491
p=0.0342
p=0.2147
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TIA1 | 1554890_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6028
p=0.0303
p=0.5197
p=0.8746
SmallAirway2
LargeAirway2
Trachea
Macrophage2

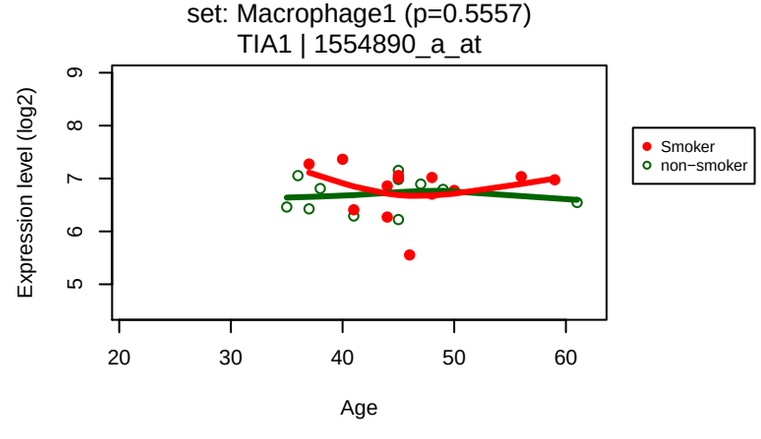

set: SmallAirway1 (p=0.2839)
TIA1 | 1554890_a_at
Expression level (log2)
Age
Smoker
non−smoker

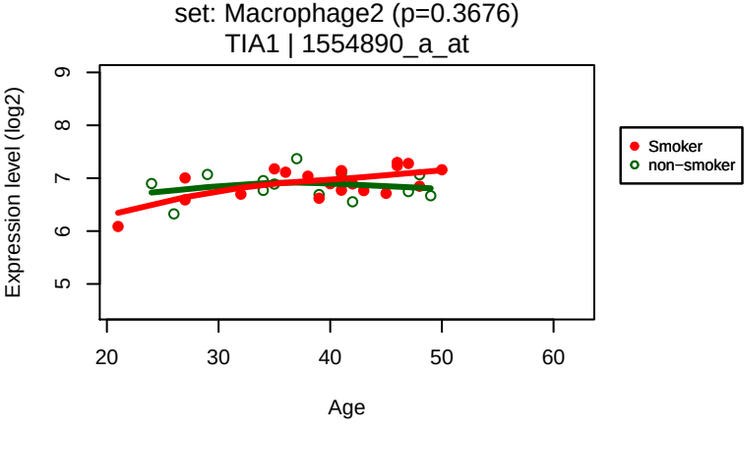

set: SmallAirway2 (p=0.6893)
TIA1 | 1554890_a_at
Expression level (log2)
Age
Smoker
non−smoker

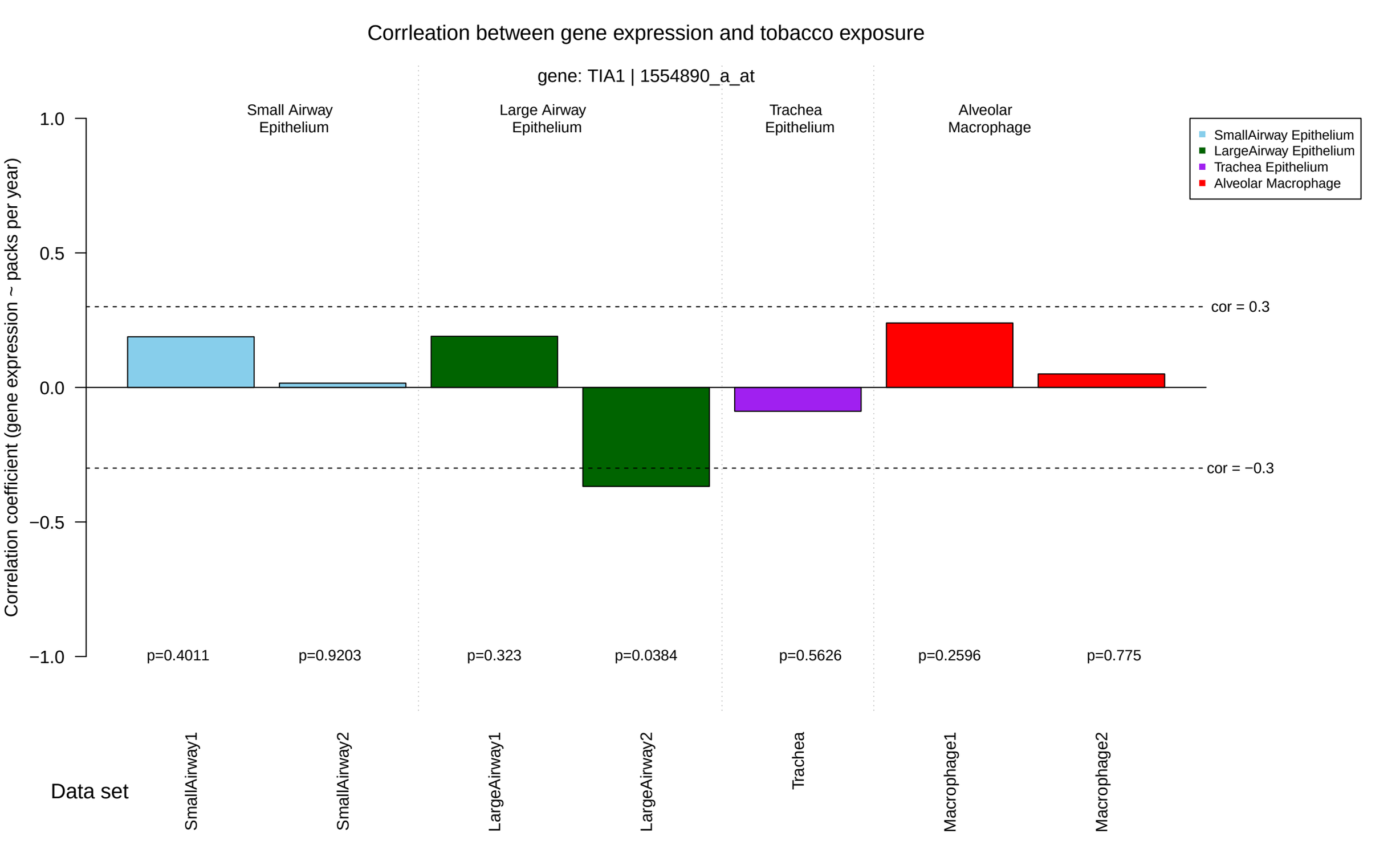

set: LargeAirway1 (p=0.8228)
TIA1 | 1554890_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.043)
TIA1 | 1554890_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1652)
TIA1 | 1554890_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5557)
TIA1 | 1554890_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3676)
TIA1 | 1554890_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TIA1 | 1554890_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4011
p=0.9203
p=0.323
p=0.0384
p=0.5626
p=0.2596
p=0.775
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2